Master of Business Analytics

Prescriptive Analytics & Optimization

MBAN 5540

Class 01
Mathematical Modeling

Instructor: Majid Taghavi

Sobey School of Business
Saint Mary's University

Prescriptive Analytics

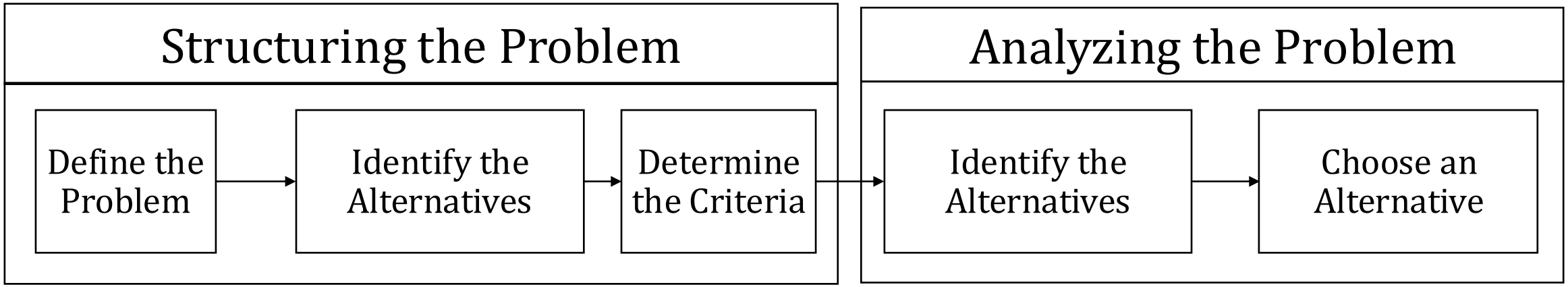
- Prescriptive analytics is all about answering this question:

"What should we do?"

How do we solve problems?

1. Define the problem.
2. Determine the set of alternative solutions.
3. Determine the criteria for evaluating alternatives.
4. Evaluate the alternatives.
5. Choose an alternative (make a decision).
6. Implement the selected alternative.
7. Evaluate the results.

Decision-making Process



Body of Knowledge

The body of knowledge involving quantitative approaches to decision making is referred to as

- Management Science
- Operations Research
- Decision Science

It had its early roots in World War II and is flourishing in business and industry due, in part, to:

- numerous methodological developments
- a virtual explosion in computing power

Why Do We Need a Quantitative Approach?

The problem is complex.

The problem is very important.

The problem is new.

The problem is repetitive.

Model Definition

Models are representations of real objects or situations

Model Definition:

A set of relationships and logical assumptions

A representation of real-world problems or phenomenon

Model Development

- Generally, experimenting with models (compared to experimenting with the real situation):
  - Requires less time
  - Is less expensive
  - Involves less risk
- The more closely the model represents the real situation, the more accurate the conclusions and predictions will be.

Mathematical Modeling

Mathematical modeling refers to the process of creating a mathematical representation of a real-world scenario to make a prediction or provide insight.

Mathematical modelling is the process of describing a real-world problem in mathematical terms, usually in the form of equations, and then using these equations both to help understand the original problem, and also to discover new features about the problem.

A mathematical model is a description of a system using mathematical concepts and language. The process of developing a mathematical model is termed mathematical modeling.

Applications

- Mathematical models are used in
  - Natural sciences: physics, biology, earth science, chemistry
  - Engineering disciplines: computer science, electrical engineering, etc.
  - Social sciences: economics, psychology, sociology, political science
  - Others: music, linguistics, and philosophy

- The main applications: solving problems in business and military

Mathematical Modeling Methods

- Linear Programming
- Integer Programming
- Non-linear Programming
- Dynamic Programming
- Monte-Carlo Simulation
- Network Programming
- Goal Programming
- Game Theory
- And many more …

Mathematical Modelling: An Example

- Break-even point is the point at which total cost and total revenue are equal, i.e., the profit is ZERO.

- Break even point (BEP)
  - Revenue = Cost
  - Profit = Revenue – Cost = 0

- To find BEP, we solve the below equation:

$$\text{Profit} = 0$$

Optimization

The process of finding the optimum (maximum or minimum) value of a **function** relative to some **set**.

The function: enables us to compare different choices

Set: represents the range of available choices

Optimization Examples

If we manage a grocery store, we try to:

- Maximize the profit
- Minimize the cost

- Minimize the # of open cash registers
- Maximize the variety and # of items
- Minimize the shelves which are empty
- Minimize the electricity charge
- Minimize the distance between shelves

More on Optimization

- The process of finding the best value (Minimum/Maximum) for an objective **function** by changing decision variables with respect to some constraints (**set**).

- It is important to identify:
  1. Decision variables
  2. Objective function
  3. Constraints

Linear Programming

- Linear programming (LP) is a form of Mathematical Programming.

- Linear programming (LP) is constructing a set of mathematical relationships to model real-world problems.

- Linear programming (LP) is an optimization technique. Thus, we need to define:
  - Decision variables
  - Objective function
  - Constraints

LP: Decision Variables

- Represent the unknown elements that we are solving for the problem; decision choices that are <u>under control,</u> such as the number of units of a product to produce.

Production Example

- Suppose x denotes the number of units produced and sold each week.

LP: Objective Function

- The goal of the problem, such as maximizing profit or minimizing cost.

Production Example

- Suppose x denotes the number of units produced and sold each week.
- The firm's objective is to maximize total weekly profit.
- With a profit of \$40 per unit, the objective function is $40x$.

LP: Constraints

- A set of restrictions or limitations that prevent us for selecting any value we please for the decision variables.

Production Example

- Suppose x denotes the number of units produced and sold each week.
- Objective function: Max $40x$
- 3 hours are required to produce each unit and only 50 hours are available per week.
- The production capacity constraint is given by $3x \leq 50$

Mathematical Model (LP)

A complete LP model for our simple production problem is:

$$\begin{array}{lll} \text{Maximize} & 40x & \text{(objective function)} \\ \text{subject to:} & 3x \leq 50 & \text{(constraint)} \\ & x \geq 0 & \text{(constraint)} \end{array}$$

- The second constraint reflects the fact that it is not possible to manufacture a negative number of units.
  - This is called "non-negativity" constraint.

Why Linear?

- Objective and constraints in *LP* problems must be expressed in terms of *linear* equations or inequalities such as:

$$2x + 5y = 10$$

- This equation is not linear: $2x^2 + 5y^3 + 3xy = 10$

- Signs of non-linearity:

$$\log(x) \qquad 2^x \qquad \sqrt{x} \qquad \frac{x}{y} \qquad xy \qquad \max(x, y)$$

- LP uses, in many cases, inequalities like:

$$x + y \leq z \quad \text{or} \quad x + y \geq z$$

Formulating a LP Problem

- Process of translating problem into LP model

- A common LP application is product mix problem.
  - Two or more products are usually produced using limited resources – such as personnel, machines, raw materials, and so on.

- Firm would like to determine:
  - How many units of each product it should produce ?
  - Maximize overall profit given its limited resources.

Example of Product Mix Problem

Flair Furniture Company produces tables and chairs:

- Each table produced results in $7 profit, each chair produced results in $5 profit.
- Each table requires: 3 hours of carpentry and 2 hours of painting.
- Each chair requires: 4 hours of carpentry and 1 hour of painting.
- Available production capacity: 2,400 hours of carpentry time and 1,000 hours of painting time.
- The marketing department wants Flair to make no more than 450 chairs.
- The marketing department wants Flair to make at least 100 tables.

Flair Furniture's problem:

- Determine best possible combination of tables and chairs to manufacture in order to attain maximum profit.

Steps in Formulating a LP Problem

- Step 1 : Define the decision variables
- Step 2 : Define the objective function
- Step 3 : Define the constraints

Step 1: Decision Variables

- The decision variables are the unknown elements of the model, and their values determine the solution of the model.

- Problem facing Flair is to determine how many chairs and tables to produce to yield maximum profit?

- Decision Variables:

T: Number of tables produced
C: Number of chairs produced

Step 2: Objective Function

- Objective function expresses the goal of the problem: Maximize profit!
- A LP model must have a <u>single</u> objective function.

- In Flair's problem, total profit may be expressed as:

($7 profit/table) (# of tables produced)
+ ($5 profit/chair)(# of chairs produced)

$$Maximize\ \ 7T + 5C$$

Step 3: Constraints

- Denote conditions that prevent one from selecting any specific subjective value for decision variables.
  - Each table requires: 3 hours of carpentry and 2 hours of painting.
  - Each chair requires: 4 hours of carpentry and 1 hour of painting.

- Constraint 1:
  - There are 2,400 carpentry hours available.

$$3T + 4C \leq 2{,}400$$

Step 3: Constraints

- Denote conditions that prevent one from selecting any specific subjective value for decision variables.
  - Each table requires: 3 hours of carpentry and 2 hours of painting.
  - Each chair requires: 4 hours of carpentry and 1 hour of painting.

- Constraint 2:
  - There are 1,000 painting hours available.

$$2T + 1C \leq 1{,}000$$

Step 3: Constraints

- Denote conditions that prevent one from selecting any specific subjective value for decision variables.

- Constraint 3:
  - There is a limit on the production of chairs (no more than 450 chairs).

$$C \leq 450$$

Step 3: Constraints

- Denote conditions that prevent one from selecting any specific subjective value for decision variables.

- Constraint 4:
  - There is a limit on the production of tables (at least 100 tables).

$$T \geq 100$$

Step 3: Constraints

- Constraint 5:
  - The **non-negativity** constraints (we can not produce negative quantities of tables and chairs).

$$T \geq 0$$

$$C \geq 0$$

Step 3: All Constraints

- Carpentry hour: $3T + 4C \leq 2{,}400$
- Painting hour: $2T + 1C \leq 1{,}000$
- Maximum chair: $C \leq 450$
- Minimum table: $T \geq 100$
- Non-negativity: $T \geq 0$
  $C \geq 0$

Flair Company LP Model

$$Maximize \quad 7T + 5C$$

Objective Function

Subject to:

Constraints

$$\begin{aligned} 3T + 4C &\leq 2{,}400 \\ 2T + 1C &\leq 1{,}000 \\ C &\leq 450 \\ T &\geq 100 \\ T &\geq 0 \\ C &\geq 0 \end{aligned}$$

Non-negativity Constraints

Decision Variables

T: Number of tables produced
C: Number of chairs produced

Business Applications

$$\text{Min} \sum_{n \in \mathcal{T}} p_n \left[\sum_{(i,j) \in \mathcal{A}} (c^u_{ij} x_{nij} + c^{up}_{nij} u^p_{nij} + c^{us}_{nij} u^s_{nij}) + \sum_{i \in \mathcal{N}} (c^{vp}_{ni} v^p_{ni} + c^{vs}_{ni} v^s_{ni}) \right]$$

$$\text{s.t.} \sum_{(j,i) \in \delta^-_i} x_{nji} - \sum_{(i,j) \in \delta^+_i} x_{nij} = d_{ni}$$

$$x_{nij} \leq C^u_{ij} \left(\sum_{m \in \bar{\mathcal{P}}(n)} u^p_{mij} + u^s_{nij} \right)$$

$$\sum_{(i,j) \in \delta^+_i} x_{nij} \leq C^v_i \left(\sum_{m \in \bar{\mathcal{P}}(n)} v^p_{mi} + v^s_{ni} \right)$$

$$x_{nij} \in \mathbb{R}^+, \; u^p_{nij}, \; u^s_{nij} \in \mathbb{Z}^+$$

$$v^p_{ni}, \; v^s_{ni} \in \mathbb{Z}^+$$